

Software Engineering Leadership

Specialization BSc Wirtschaftsinformatik / Business Information Technology

Designed for:

students interested in **advanced software development** in business information technology.

Programme focus:

- DevOps
- Software Engineering
- Software Architecture
- IT-Project

Recommended prerequisites:

Requirements Engineering

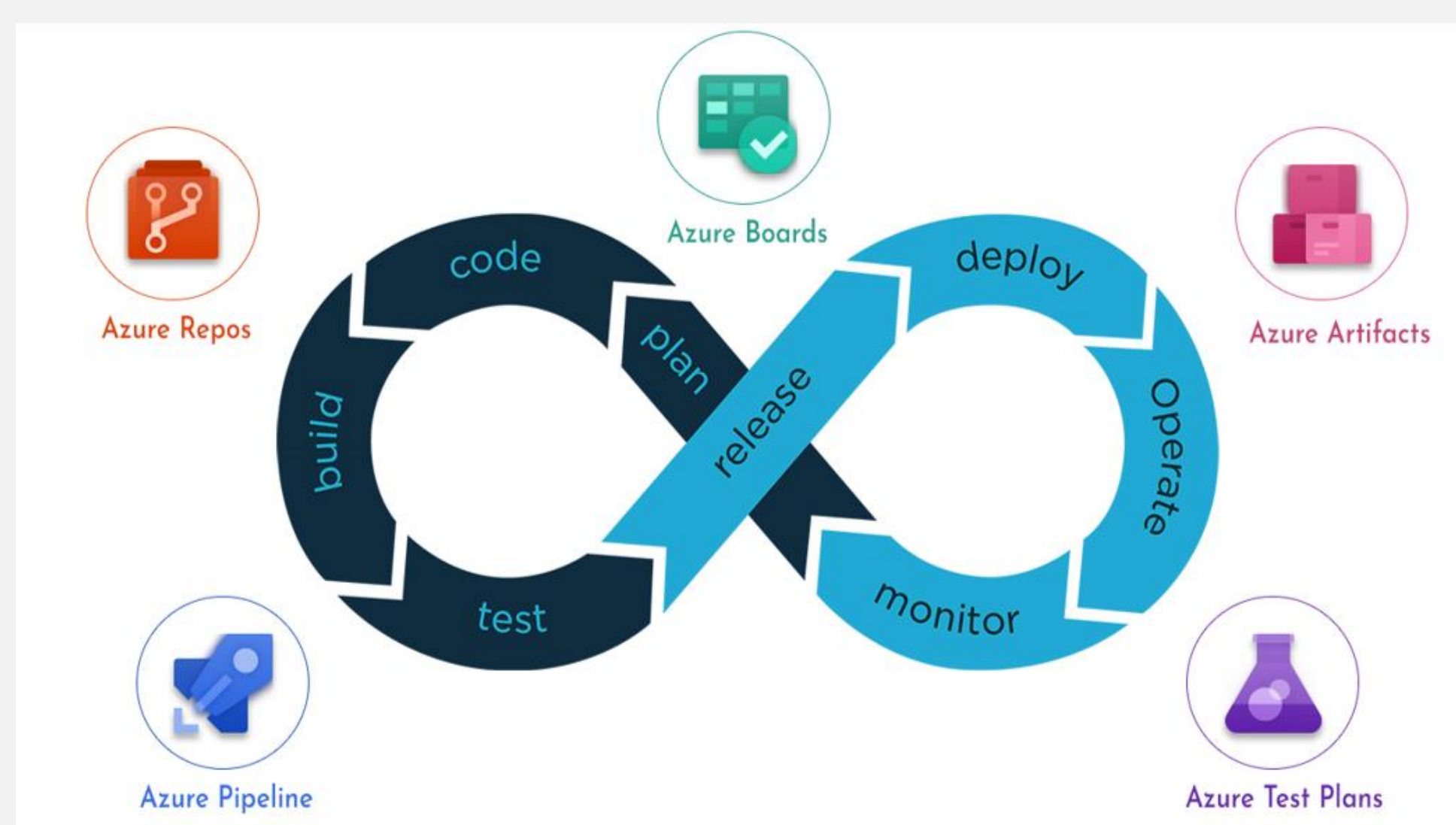
Programming 1+2

Database Technology

Internet Technology

Agile Applicaion Lifecycle Management *

- Hands-on classes with Azure DevOps to learn state-of-the-art infrastructure for agile development
- Agile Requirements Engineering (Scrum, Kanban)
- Code management with Git and GitHub
- Project building and deployment
- Test case development



Software Architecture *

- Provide a better **understanding of the architecture of business applications** as the foundation for supporting the ongoing digitalization.
 - Architectures and design principles for single applications and distributed applications.
- Provide **hands on experience** for the integration of business applications into larger business processes and workflows.
 - Cover common, well established integration styles and technologies
 - Look at recent/current developments.
- **Rough content overview:**
 - Architecture(s) for a single Applications: Layered Architecture and especially Microservices
 - Architectures and technologies for integrating multiple applications:
 - Topics from the scope of Enterprise Application Integration (theoretical and practical)
 - SOA and orchestration through services and messaging for example with an Enterprise Service Bus (ESB)
 - Enterprise Integration Patterns (EIP)
 - WSDL / Web Services / REST
 - (Complex) Event Processing

Software Engineering *

- Refresher and Foundations
 - Important aspects of real-world Java programming
- Advanced Java programming
 - Lambdas, generics, concurrency
- Components for web services
 - Implementing web services from the ground up
- Using frameworks (advantages and disadvantages)
 - Practical project with Spring

IT-Project

- Group work (3 – 5 Personen)
- Realisation of a small software project
 - Businesscase oder game
 - Client-server application with Spring
- Different project phases divided over the semester
 - Requirements
 - Technical specification
 - Implementation
 - Testing
 - Final discussions

After graduation:

DevOps Engineer

You are responsible for the design of processes and environments that are necessary during the entire life cycle of a software.

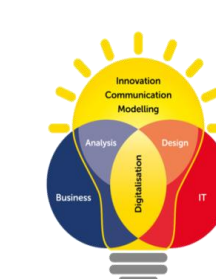
Junior (Java) Developer

You have the necessary skills to apply tools for software development in a company and to work productively on projects.

Application Manager

You are responsible for the ongoing operation of an application, coordinating incidents, release installations and change requests.

* Can also be chosen as an elective module



member of
swissuniversities

